



Multi-task encrypted network traffic classification based on feature extraction

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Abstract

Aiming to address the issue of traditional encrypted network traffic identification models' shortcomings in feature extraction and the use of separate models for traffic characterization and application identification in encrypted traffic classification, which makes network management more difficult. Here, a Transformer-1DCNN-ECA-based multi-task encrypted traffic categorization model is suggested. First, the dataset is cut, filled, and normalized to increase the classification accuracy; next, the Transformer model's multi-head attention mechanism is used to capture long-distance feature dependencies and finish extracting global feature information; finally, 1DCNN is used to extract local features; Lastly, the network becomes more focused on important features with the addition of a lightweight ECA module, which enhances model performance and enables multi-task encrypted traffic categorization. The average accuracy of this model on ISCX VPN-nonVPN dataset for traffic feature and application identification classification is 97.97% and 98.27%, which is better than other models.

CCS Concepts

• **Security and privacy** → Human and societal aspects of security and privacy; Usability in security and privacy.

Keywords

Encrypted network traffic classification, Multitasking, Feature extraction, Transformer, 1DCNN, ECA

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1 INTRODUCTION

Some network administrators are confronted with the challenge of classifying and handling encrypted traffic as its use grows in popularity. Conventional techniques based on port number and protocol type typically rely on the protocol's plaintext information

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or the designated port number and employ rules or feature sets to differentiate between various forms of network traffic. As a result, this approach has a low classification accuracy and is unable to identify complex protocols and extremely covert attack traffic. Two of the more important responsibilities in network traffic categorization are traffic characterization and application identification. The majority of current solutions for these two tasks typically use a single-task processing paradigm. While this method can partially satisfy the categorization requirements, it tends to lead to an increase in the complexity of network management in practice.

This research suggests a multi-task encrypted network traffic categorization approach based on Transformer-1DCNN-ECA to overcome the aforementioned issues. The model increases the effectiveness of encrypted traffic categorization and allows simultaneous multi-task learning by enabling application identification and traffic characterization on the ISCX VPN-nonVON dataset.

2 RELATED WORKS

Deep learning algorithms have improved results in computer vision, image processing, and other fields due to the rapid development of artificial intelligence and big data. Researchers studying encrypted network traffic have also started working with encrypted traffic data that has various representations and have started using deep learning models to automatically extract features to achieve detection and classification. Although it circumvents the drawbacks of conventional DPI-based techniques for handling scenarios like packet ordering, Dias et al.'s^[1] novel approach for traffic categorization and identification using the ML method still has certain restrictions. This approach combines K-means and KNN. A traffic classification model based on LSTM and BERT was proposed by Shi et al.^[2] It achieves good classification accuracy, but the packet class imbalance issue is still a common one that could cause the model's performance to deteriorate. Lin et al.^[3] proposed a Bidirectional Encoder Representation (BERT) for Convolutional Neural Network Transformers. This classification model improves the accuracy of classification but at a higher computational cost and time spent.

While multi-task classification models have not received much attention, the aforementioned issues emerge for single-task encrypted traffic categorization. To achieve more accurate and efficient multi-task encrypted traffic classification, this paper suggests a deep learning model-based encrypted traffic classification by examining the issues and models of single-task and multi-task encrypted traffic classification. The model is comparatively faster and less costly to compute, and it uses Transformer-1DCNN-ECA to obtain feature information.

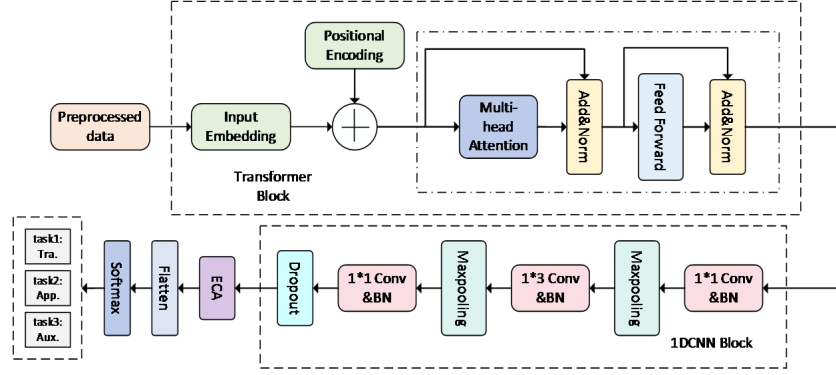


Figure 1: The model's general framework.

3 CLASSIFICATION METHODOLOGY OF THIS PAPER

3.1 Modeling Framework

The general framework diagram for the Transformer-1DCNN-ECA multi-task encrypted traffic classification model presented in this paper is displayed in Figure 1. To achieve the extraction of global information, the raw encrypted traffic data is first used as the model input following preprocessing; second, the Transformer encoder's input is subjected to a summation operation using word list coding and position coding; the result is then input into the multi-head attention layer and produced through the feed-forward network, where residual connectivity and layer normalization operations are performed between the layers; The output data is once more subjected to a one-dimensional convolutional layer for the extraction of local features, followed by the ECA attention module for the extraction of key feature information, the Flatten layer for the one-dimensionalization of the multidimensional data, the Softmax classifier for the multi-task encrypted traffic classification, and the output of the results.

3.2 Data Pre-processing

(1) The Ethernet header is first eliminated during the preprocessing stage because the ISCX VPN-nonVPN dataset is recorded at the data link layer and the physical link information it contains is useless for identifying applications or traffic types and may lead to model overfitting issues. Then the IP address in the IP header is changed to 0.0.0.0;

(2) The two protocols, TCP and UDP, therefore have different header lengths in the transport layer; the TCP protocol typically has a header length of 20 bytes, while the UDP protocol has a header length of 8 bytes. To make the lengths equal, you must add 0 at the end of the UDP protocol header;

(3) Furthermore, Packets that execute the three-way TCP handshake and exclude any substantive data, such as DNS, ACK, SYN, and FIN packets, predominantly utilized in service protocols, were removed.

(4) Lastly, 0 bytes to 1500 bytes are added to the end of each packet to guarantee consistency of the model input data. Each byte value is divided by 255 to ensure the input values are normalized

and fall within the range of $[0, 1]$. The model's initial input data format is the 1500-byte sequence that has been handled in this way.

3.3 Transformer Encoder Block

After preprocessing, the traffic data is fed into the Transformer encoder^[4]. The Transformer model encodes the input data, which is a sequence. A positional encoding layer is added to address the issue of positional information in the sequence by appending a corresponding positional vector to each position. The following formula is used to determine the position encoding.

$$\begin{cases} p_i^{(2s)} = \sin\left(i/10000^{2s/d}\right) \\ p_{i+1}^{(2s)} = \cos\left(i/10000^{2s/d}\right) \end{cases} \quad (1)$$

where i is the time step; s is the number of dimensions; and d denotes the dimensionality of the model.

The encoder and decoder are the two components that make up the Transformer's structure. In this work, we use the encoder, which is mostly utilized for feature extraction. A crucial component of the feature extraction method is the Multi-Head Attention layer, which allows each attention head to acquire distinct weights and capture information in several representation subspaces. The single-head self-attention module, which maps the input sequence into a query matrix Q , a key-value matrix K , and a value matrix V , where Q, K , and $V \in R^{n \times d_k}$, and then linearly transforms and connects the outputs of multiple heads to realize the extraction of global feature information. The corresponding computational formulas are displayed below.

$$\text{Attention} = \text{soft max} \left(\frac{QK^T}{\sqrt{d_k}} \right) V \quad (2)$$

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{Head}_1, \text{Head}_2, \dots, \text{Head}_h) W \quad (3)$$

$$\text{Head}_i = \text{Attention} \left(QW_i^Q, KW_i^K, VW_i^V \right) \quad (4)$$

where d_k denotes the channel dimension and $\sqrt{d_k}$ is the scaling factor. In this paper, we choose $h = 6$, i.e., it consists of 6 self-attention mechanisms.

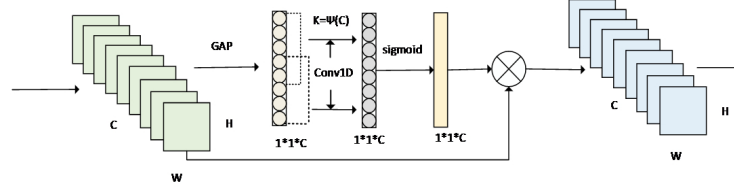


Figure 2: ECA structure diagram.

Table 1: Traffic Type Label, Application Label and Auxiliary Task Label Information.

task1: Traffic type	task2: Application	task3: Auxiliary task
Email,File Transfer, Chat,Torrent,Streaming,Voip, VPN:Email,VPN:Chat,VPN:Streaming, VPN:File Transfer, VPN:Voip,VPN:Torrent	AIM Chat,Facebook,Email, Netflix,SCP,Skype,SFTP, FTPS,ICQ,Hangouts,Gmail, Youtube,Voipbuster, Spotify,Torrent,Vimeo,Tor	All_file Transfer, All_Email,All_Chat All_voip,All_Torrent, All_Streaming

3.4 1DCNN Block

Even though the encrypted traffic data information has undergone global feature extraction after going via the Transformer encoding module, it may still have relatively low classification accuracy if it is produced directly. To compensate for this and carry out more fine-grained feature extraction, the CNN module—which is more sensitive to local feature extraction—must be added. Since encrypted communication is regarded as sequential data and the 1DCNN module is more suited for traffic categorization than 2DCNN, 1DCNN was selected in this instance.

Three convolutional layers, two maximum pooling layers, and one dropout layer make up the 1DCNN module. Each convolutional layer has a batch normalization layer to normalize the data, which can effectively prevent gradient vanishing and gradient explosion and increase the network’s stability. With a value of 0.5, the final Dropout layer can be somewhat more successful in reducing overfitting to produce the regularization effect.

3.5 ECA Block

Following the extraction of data information using 1DCNN, an effective and lightweight ECA is introduced to reduce the computational complexity and number of parameters, improve the network’s ability to pay attention to the encrypted traffic data, and highlight the important features of the traffic data while suppressing redundant information by adaptively adjusting the weights of each channel. In Figure 2, the ECA structure is displayed.

4 EXPERIMENT AND RESULT ANALYSIS

4.1 Experimental Environment

The software framework used for the experimental platform is Python 3.12, Cuda 12.1, and the experimental model training library is PyTorch 2.3.0, running under the ubuntu22.04 system environment, with 48GB of RAM, an Intel(R) Xeon (R) Platinum 8352 CPU, and an RTX 3080x2 GPU (20GB).

4.2 Experiment Dataset

The publicly available ISCX VPN-nonVPN encrypted traffic dataset is chosen for the experiments, in multi-task encrypted traffic categorization, here the dataset will be reclassified into 12 traffic type labels and 17 application labels, at the same time, a simple auxiliary task can improve its categorization accuracy when performing a complex task, therefore, an auxiliary task is introduced here, which does not distinguish between encrypted traffic and normal traffic, but only categorized by type. Table 1 displays the labeling data for the three tasks.

4.3 Evaluation Metrics

In statistical classification, four often used measures are precision, accuracy, F1-score and recall . Here, accuracy, F1-score, and recall are selected as assessment criteria for traffic categorization to more thoroughly assess the Transformer-1DCNN-ECA multi-tasking model. The following formula is used to determine these metrics.

$$\text{Accuracy} = \frac{\text{TN} + \text{TP}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}} \quad (5)$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (6)$$

$$\text{Recall} = \frac{\text{TP}}{\text{FN} + \text{TP}} \quad (7)$$

$$\text{F1} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (8)$$

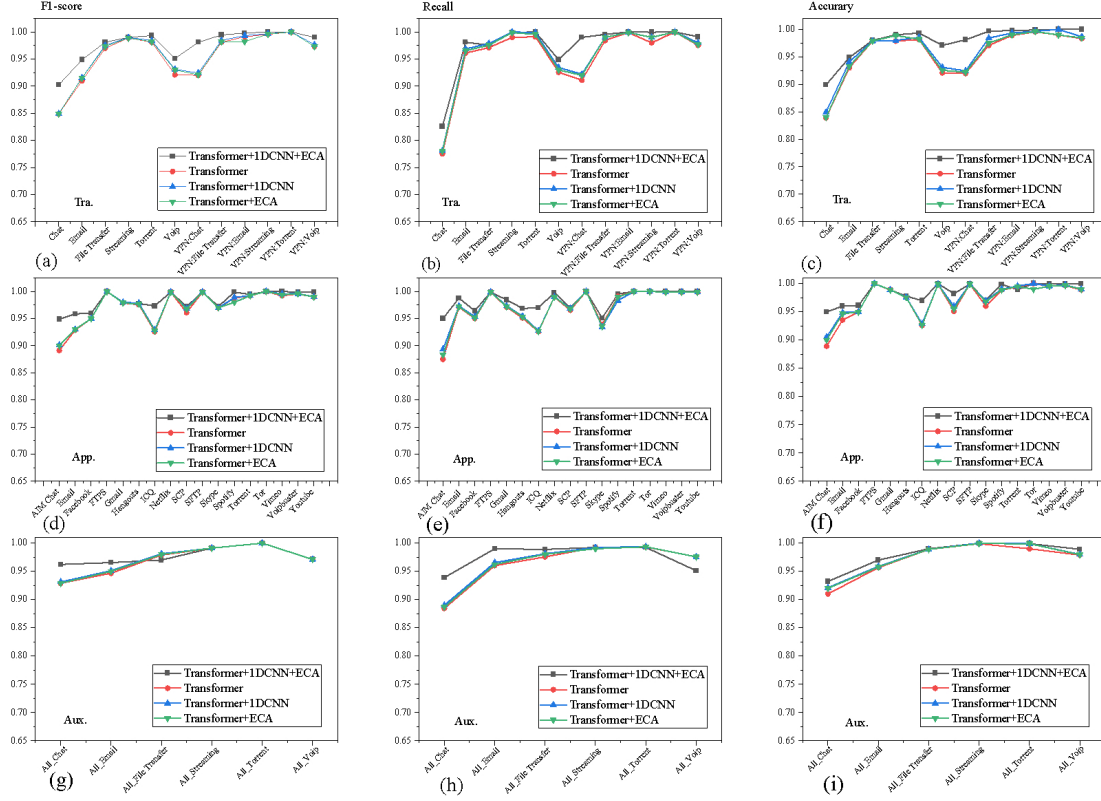
where TP represents true positive, FP signifies false positive, TN stands for true negative, and FN denotes false negative.

4.4 Analysis of Experimental Results

4.4.1 Ablation Experiment. The model architecture of this paper is Transformer-1DCNN-ECA, and The Transformer model^[4] is used as a baseline model to analyze the role of different modules by comparing other models. The results are shown in Table 2 and Figure 3.

Table 2: Ablation experiment results for each model for classification on auxiliary tasks, application identification, and flow characterization.

Model	Tra.			App.			Aux.		
	Acc	F1	Recall	Acc	F1	Recall	Acc	F1	Recall
Baseline	0.9599	0.9590	0.9544	0.9727	0.9727	0.9718	0.9689	0.9686	0.9650
+1DCNN	0.9618	0.9610	0.9619	0.9749	0.9745	0.9758	0.9713	0.9710	0.9699
+ECA	0.9657	0.9655	0.9611	0.9748	0.9744	0.9755	0.9714	0.9711	0.9697
+1DCNN+ECA	0.9797	0.9796	0.9760	0.9827	0.9824	0.9832	0.9790	0.9789	0.9780

**Figure 3: (a) (b) (c) (d) (e) (f) (g) (g) (h) (i) are F1-score, Recall and Accuracy classification plots of each model on each categories of traffic characteristics, application identification and auxiliary tasks, respectively**

According to the findings of the ablation experiment, the suggested model outperforms the benchmark model by roughly 2% in terms of accuracy, F1-score, and recall. Thus, this paper’s suggested model can fully utilize the Transformer and 1D-CNN block features to extract information related to long distance from traffic data, extract local detail information, and then use the ECA features for multi-task classification of encrypted traffic.

4.4.2 Comparison Experiments. The model in this research is compared with six related working models for comparative studies to further validate the model’s efficacy. Table 3 displays the findings of the comparison tests of F1-score and Recall with six other popular deep learning models on the dataset for 12 traffic-type classifications and 17 application-type classifications.

As suggested in this research, the model outperforms other popular models on both tasks, as the above table demonstrates.

5 CONCLUSION

This paper establishes a multi-task encrypted traffic classification model based on Transformer-1DCNN-ECA to address the issues of inadequate extracted feature information and complex network traffic management of traditional traffic classification methods. The model’s performance for multi-task encrypted traffic classification is improved by first using the Transformer model’s coding part to extract global features, then using 1DCNN for the extracted global features to achieve a fine-grained portrayal of local features, and finally using the ECA model for important feature extraction. By

Table 3: Accuracy, F1-score, and Recall Results Comparison for Various Models.

Methods on ISCX VPN-non VPN	Tra.			App.		
	Acc	F1	Recall	Acc	F1	Recall
1DCNN ^[5]	0.9329	0.9377	0.9306	0.9758	0.9785	0.9745
SAE ^[5]	0.9200	0.9200	0.9200	0.9500	0.9500	0.9500
MTC ^[6]	0.9787	0.9794	0.9754	0.9825	0.9825	0.9830
TSCRNN ^[7]	-	0.9260	0.9260	-	-	-
CNN+GRU ^[8]	-	-	-	0.9611	0.9608	-
RBRN ^[9]	0.9722	0.9721	0.9733	-	-	-
Transformer-1DCNN-ECA	0.9797	0.9796	0.9760	0.9827	0.9824	0.9832

conducting ablation studies on the ISCX VPN-nonVPN dataset, the influence of each sub-module within the model on the experimental results is elucidated, and it is also illustrated that the proposed model exhibits superior classification capabilities. In conclusion, comparative experiments reveal that the proposed model surpasses other established related works in terms of Recall, Accuracy and F1-score metrics on the dataset.

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